

**MASTER IN ARTIFICIAL INTELLIGENCE
APPLIED TO SPORTS**

Module 9

**The great challenge:
master's final project**

Final Project Proposals



Final Project Ideas

The following proposals represent potential directions for your Master's final project.

They are grounded in real challenges and opportunities currently emerging across the sports and technology industry.

These ideas reflect how AI is reshaping decision-making, performance, and innovation in sport.

More than academic exercises, they are opportunities to build solutions with real-world relevance and impact.

Your project should not just demonstrate technical skills — it should position you in the market by solving a meaningful problem.

AI-Driven Optimization of Sports Training

Objective

Develop an AI system to analyze athlete performance data and personalize training programs. Include movement analysis, health monitoring, and AI-based recommendations to maximize performance and reduce injury risk.

Suggested Approach

- Collect data (performance, workload, physiological metrics)
- Preprocess and explore patterns
- Build models for performance and injury prediction
- Generate personalized training recommendations
- Visualize insights through dashboards or apps

AI Areas

- Machine Learning
- Deep Learning
- Computer Vision
- Time Series Analysis
- Recommender Systems
- Anomaly Detection

Sports Outcome Prediction

Objective

Develop a predictive model using historical data and real-time statistics to forecast sports event outcomes. Focus on a specific sport and include variables such as team performance, individual player statistics, weather conditions, and more.

Suggested Approach

- Collect historical and real-time data
- Select key features (team, players, external factors)
- Train predictive models
- Evaluate and optimize model performance
- Present predictions through dashboards or visual tools

AI Areas

- Machine Learning
- Time Series Analysis
- Predictive Analytics
- Feature Engineering
- Ensemble Methods

Sentiment Analysis on Social Media for Sports Brand Management

Objective

Develop a Natural Language Processing (NLP) system to analyze fan sentiment on social media regarding teams, players, and sports brands.

Provide organizations with valuable insights into public perception to improve marketing and communication strategies.

Suggested Approach

- Collect social media data (Twitter, Instagram, forums)
- Clean and preprocess text data
- Apply sentiment analysis models
- Extract key topics and trends
- Visualize insights through dashboards

AI Areas

- Natural Language Processing (NLP)
- Sentiment Analysis
- Text Mining
- Topic Modeling
- Deep Learning (Transformers)

AI-Powered Virtual Assistant for Sports Training and Health

Objective

Develop an AI-based virtual assistant that provides personalized advice on training, nutrition, and recovery for athletes.

The system can integrate with wearables to monitor health and performance in real time.

Suggested Approach

- Collect user and wearable data (health, activity, performance)
- Build conversational interface (chatbot/voice assistant)
- Develop recommendation system for training and nutrition
- Integrate real-time monitoring and feedback
- Deploy as mobile app or web platform

AI Areas

- Natural Language Processing (NLP)
- Conversational AI (Chatbots)
- Recommender Systems
- Time Series Analysis
- Wearable Data Analytics

Predictive Analytics for Ticket Sales and Merchandising Optimization

Objective

Use predictive analytics techniques to optimize ticket sales and merchandising by analyzing fan purchasing patterns and preferences.

Help clubs maximize revenue and improve fan satisfaction.

Suggested Approach

- Collect sales and customer data (tickets, merchandise, behavior)
- Analyze purchasing patterns and segmentation
- Build predictive models for demand forecasting
- Develop recommendation systems for products and offers
- Visualize insights through dashboards

AI Areas

- Machine Learning
- Predictive Analytics
- Customer Segmentation
- Recommender Systems
- Time Series Forecasting

Game Analysis and Sports Strategy with AI

Objective

Develop a system to analyze games and sports tactics, providing coaches and analysts with advanced insights into game strategy.

Use video analysis and real-time tracking data to evaluate the effectiveness of different strategies and tactics.

Suggested Approach

- Collect video and tracking data
- Apply computer vision for player and ball tracking
- Analyze tactical patterns and formations
- Evaluate strategy effectiveness using data
- Visualize insights for coaches and analysts

AI Areas

- Computer Vision
- Machine Learning
- Deep Learning
- Spatiotemporal Analysis
- Pattern Recognition

Talent Detection and Analysis in Sports with AI

Objective

Develop an AI tool to identify and analyze young sports talents by evaluating performance, technical skills, and progression.

Include talent profiling and prediction of future development.

Suggested Approach

- Collect performance and player development data
- Define key metrics for talent identification
- Build models to evaluate and rank players
- Predict future potential and progression
- Visualize player profiles and insights

AI Areas

- Machine Learning
- Predictive Analytics
- Player Profiling
- Time Series Analysis
- Data Mining

Advanced Performance Metrics Generation

Objective

Develop a system that uses real-time performance data to create advanced, customized metrics for players and teams across different sports.

Enable comparisons across leagues and competition levels.

Suggested Approach

- Collect real-time performance data
- Define and engineer advanced metrics
- Build models to generate and validate metrics
- Enable cross-league and cross-level comparisons
- Visualize insights for analysts and coaches

AI Areas

- Machine Learning
- Feature Engineering
- Data Analytics
- Time Series Analysis
- Statistical Modeling

Adaptability of Metrics Across Leagues with AI

Objective

Develop an AI model capable of adapting performance metrics and analysis to different leagues and sports.

Consider factors such as playing style, specific rules, and competition level.

Suggested Approach

- Collect multi-league and multi-sport data
- Identify key differences (rules, pace, style)
- Normalize and standardize metrics
- Build models to adjust and compare performance
- Create benchmarking tools across leagues

AI Areas

- Machine Learning
- Feature Engineering
- Domain Adaptation
- Statistical Modeling
- Data Normalization

Computer Vision Applications for Sports Analysis

Objective

Use computer vision techniques to analyze match and training videos, extracting data on player positions, team tactics, and key game events.

Support pattern detection and analysis of athletes' physical and technical performance.

Suggested Approach

- Collect and process video data
- Apply player and object detection models
- Track player movements and positions
- Identify patterns and key events
- Visualize insights for performance analysis

AI Areas

- Computer Vision
- Deep Learning
- Object Detection & Tracking
- Spatiotemporal Analysis
- Pattern Recognition

Athlete Salary Estimation with AI Models

Objective

Develop a predictive model to estimate athlete salaries based on performance, popularity, team impact, and current market conditions.

Incorporate data from multiple sources, including social media, match performance, and historical contract data.

Suggested Approach

- Collect multi-source data (performance, contracts, media presence)
- Define key features influencing salary
- Train regression models for salary prediction
- Evaluate model accuracy and fairness
- Visualize insights for decision-making

AI Areas

- Machine Learning (Regression Models)
- Predictive Analytics
- Feature Engineering
- Data Integration
- Econometric Modeling

Player Transfer Value Estimation with AI

Objective

Develop an AI system to estimate player market value by analyzing multiple factors such as on-field performance, market potential, demand, and contract situation.
Support clubs in making better decisions in player transfers.

Suggested Approach

- Collect multi-source data (performance, contracts, market trends)
- Define key variables influencing player value
- Train predictive models for valuation
- Validate model accuracy with historical transfers
- Visualize insights for decision-making

AI Areas

- Machine Learning (Regression Models)
- Predictive Analytics
- Feature Engineering
- Market Modeling
- Data Integration

Injury Prevention through Movement Asymmetry Analysis with AI

Objective

Use computer vision and machine learning to analyze training and competition videos, identifying movement asymmetries and patterns that may indicate a high risk of injury.
Integrate wearable data for deeper analysis.

Suggested Approach

- Collect video and wearable data
- Apply pose estimation and movement tracking
- Detect asymmetries and abnormal patterns
- Build models to predict injury risk
- Provide actionable insights for training adjustments

AI Areas

- Computer Vision
- Machine Learning
- Pose Estimation
- Time Series Analysis
- Biomechanical Analysis

Analysis of Playing Style Adaptability to Different Coaches with AI

Objective

Develop an AI-based system to evaluate how players and team tactics adapt to different coaching styles and philosophies.

Suggested Approach

- Collect tactical, performance, and player data
- Model different coaching styles and playing philosophies
- Analyze team and player adaptability
- Simulate tactical changes and their impact
- Provide insights for squad optimization

AI Areas

- Machine Learning
- Pattern Recognition
- Tactical Modeling
- Simulation & Predictive Modeling
- Data Analytics

Natural Language Processing Applications in Sports

Objective

Implement NLP models to analyze match commentary, interviews with players and coaches, and social media posts.

Extract insights on team perception, player sentiment, and trends in public sports discussions.

Suggested Approach

- Collect text data (media, interviews, social platforms)
- Clean and preprocess textual data
- Apply NLP models for sentiment and topic analysis
- Detect trends and key insights
- Visualize results for decision-making

AI Areas

- Natural Language Processing (NLP)
- Sentiment Analysis
- Topic Modeling
- Text Mining
- Transformer Models

Anthropometry Analysis and Its Impact on Sports Performance

Objective

Analyze how athletes' anthropometric characteristics influence performance across different sports using AI to correlate physical measurements with performance data.
Support talent identification and personalized training and nutrition plans.

Suggested Approach

- Collect anthropometric and performance data
- Analyze correlations between physical traits and performance
- Build models to identify key success factors
- Segment athletes based on profiles
- Visualize insights for training and scouting

AI Areas

- Machine Learning
- Statistical Analysis
- Predictive Modeling
- Data Analytics
- Clustering & Segmentation

AI-Powered Strategic Career Advisory System for Athletes

Objective

Develop an AI system that supports athletes in critical career decisions, such as team transfers, by providing analysis and personalized recommendations to enhance long-term professional growth.

Suggested Approach

- Collect career, performance, and market data
- Analyze player development and team fit
- Build models to evaluate career scenarios
- Predict long-term impact of decisions
- Provide personalized recommendations

AI Areas

- Machine Learning
- Predictive Modeling
- Recommender Systems
- Decision Support Systems
- Data Analytics

Social Media Analysis for Mental and Psychological Load Assessment in Athletes

Objective

Develop an AI system to analyze players' social media activity and posts to assess their mental and psychological state. Use NLP and sentiment analysis to detect patterns that may indicate stress, mental load, or risk of mental health issues.

Suggested Approach

- Collect social media data (posts, interactions)
- Preprocess and clean textual data
- Apply sentiment and emotion analysis models
- Detect behavioral patterns and anomalies
- Build a psychological well-being index

AI Areas

- Natural Language Processing (NLP)
- Sentiment Analysis
- Emotion Detection
- Anomaly Detection
- Behavioral Analytics

Player Intelligence and Decision Quality Index with AI

Objective

Develop a system to evaluate players' decision-making and game intelligence using advanced data analysis and machine learning.

Create metrics that reflect decision quality, game vision, effectiveness in critical situations, and overall team contribution.

Suggested Approach

- Collect event and tracking data (passes, shots, movements)
- Define metrics for decision quality and context
- Build models to evaluate actions in different scenarios
- Quantify player impact beyond traditional statistics
- Visualize insights for analysts and coaches

AI Areas

- Machine Learning
- Advanced Analytics
- Feature Engineering
- Spatiotemporal Analysis
- Decision Modeling

Generative AI for Automated Match Analysis Reports

Objective

Develop a generative AI system that automatically creates match reports based on game data and events. Transform raw data into human-like narratives for coaches, analysts, and media.

Suggested Approach

- Collect match data (events, stats, tracking)
- Structure key insights and moments
- Use LLMs to generate natural language reports
- Customize tone (technical, media, fan-focused)
- Integrate into dashboards or reporting tools

AI Areas

- Generative AI
- Large Language Models (LLMs)
- Natural Language Generation (NLG)
- Data-to-Text Systems

Generative AI for Personalized Training Plan Creation

Objective

Develop a generative AI system that creates personalized training plans for athletes based on performance, goals, and physical condition.

Suggested Approach

- Collect athlete data (performance, health, goals)
- Define constraints and training principles
- Use generative models to create tailored plans
- Continuously adapt plans based on new data
- Deliver via app or virtual assistant

AI Areas

- Generative AI
- Large Language Models (LLMs)
- Recommender Systems
- Reinforcement Learning

Multi-Sport AI Agent for Performance Monitoring and Optimization

Objective

Develop an AI agent that monitors and analyzes athlete performance across different sports, adapting insights and recommendations to each sport's specific context.

Suggested Approach

- Collect multi-sport performance and tracking data
- Build an agent that understands sport-specific contexts
- Adapt metrics and benchmarks dynamically
- Provide personalized recommendations per sport
- Continuously learn from new data across disciplines

AI Areas

- AI Agents
- Multi-Domain Learning
- Machine Learning
- Time Series Analysis
- Adaptive Systems

Multi-Sport AI Agent for Talent Identification and Transferability

Objective

Develop an AI agent that identifies talent across different sports and evaluates the potential transferability of skills between disciplines.

Suggested Approach

- Collect data from multiple sports (performance, physical, technical)
- Build an agent that compares skill profiles across sports
- Identify transferable attributes (speed, coordination, decision-making)
- Predict success in alternative sports
- Generate cross-sport talent recommendations

AI Areas

- AI Agents
- Transfer Learning
- Predictive Modeling
- Clustering & Profiling
- Cross-Domain Analytics

Multi-Sport Image Processing for Automated Performance Tracking

Objective

Develop a computer vision system capable of analyzing images and videos across multiple sports to automatically track athletes, movements, and key actions.

Suggested Approach

- Collect image and video data from different sports
- Apply object detection (players, ball, equipment)
- Use pose estimation for movement analysis
- Adapt models to sport-specific contexts
- Generate performance metrics automatically

AI Areas

- Computer Vision
- Image Processing
- Object Detection
- Pose Estimation
- Deep Learning

Computer Vision for Automatic Highlight and Event Detection in Multi-Sport

Objective

Develop a system that uses computer vision to automatically detect key events (goals, shots, fouls, actions) and generate highlights across different sports.

Suggested Approach

- Collect and label multi-sport video datasets
- Detect players, ball, and key interactions
- Train models to recognize important events
- Segment and generate highlight clips automatically
- Deliver content for media, fans, and analysts

AI Areas

- Computer Vision
- Action Recognition
- Video Understanding
- Deep Learning
- Event Detection

AI-Powered Crowd and Atmosphere Analysis in Sports Events

Objective

Develop an AI system to analyze crowd noise and stadium atmosphere to measure fan engagement, emotional reactions, and event impact.

Suggested Approach

- Collect audio data from stadiums and broadcasts
- Process sound signals (volume, frequency, patterns)
- Detect key moments (cheers, boos, reactions)
- Correlate crowd response with game events
- Visualize fan engagement metrics

AI Areas

- Audio Signal Processing
- Machine Learning
- Sound Event Detection
- Pattern Recognition
- Time Series Analysis

AI-Based Engine Sound Analysis for Performance Optimization

Objective

Develop an AI system that analyzes engine sound data to monitor performance, detect anomalies, and optimize vehicle efficiency in real time.

Suggested Approach

- Collect audio data from engines during races and testing
- Process sound signals (frequency, vibration patterns)
- Train models to detect performance deviations
- Identify mechanical issues early
- Provide real-time feedback to engineers and drivers

AI Areas

- Audio Signal Processing
- Machine Learning
- Anomaly Detection
- Time Series Analysis
- Predictive Maintenance

AI-Based Multi-Sport Compatibility Analysis using Anthropometrics and Neuromuscular Response

Objective

Develop an AI system to evaluate children's compatibility with different sports by analyzing anthropometric data, physical measurements, and neuromuscular responses. Support early talent identification and optimal sport selection.

Suggested Approach

- Collect anthropometric data (height, weight, body composition)
- Measure neuromuscular responses (reaction time, strength, coordination)
- Build profiles for different sports requirements
- Train models to match athlete profiles with sport demands
- Generate personalized sport recommendations

AI Areas

- Machine Learning
- Biomechanical Analysis
- Predictive Modeling
- Clustering & Profiling
- Data Analytics

**Start building today
what you want to be
known for tomorrow.**

